

EDUCATION:

Brown University, Providence, RI | *Master of Science, Computer Science*

(Fall 2024 - May 2025)

Brown University, Providence, RI | *Bachelor of Science, Computer Science*

(Fall 2020 - May 2024)

WORK EXPERIENCE:

- **C++ Software Engineer, Dassault Systemes – Johnston, RI** (June 2025 - Present)
 - Shipped critical hotfixes and core optimizations to enterprise clients, reducing application latency and improving reliability..
 - Modularized hypervisor codebase into a lightweight portable dependency, improving cross-team development velocity.
 - Hardened CI/CD pipeline by developing E2E native app live capture automated tests with JavaScript user interaction scripting.
 - Increased automated unit, integration, and E2E test coverage by 50%+, systematically reducing workflow regressions.
 - Developed Python trace file analysis tool to identify performance gaps, accelerating root-cause analysis in production.
 - Leading design of internal AI-powered triage tooling, targeting reduced time-to-root-cause for high-frequency failure modes.
- **Fullstack Software Engineer Intern, Onset Data Loggers - LI-COR – Bourne, MA** (May 2024 - August 2024)
 - Engineered MQTT handler processing 10k+ IoT messages/day, accelerating device-cloud communication throughput.
 - Built Vue.js firmware management interface for cloud data operations, reducing manual engineering setup by 60%.
 - Prototyped a mapping dashboard with Leaflet and GIS APIs, enabling real-time user interaction with live weather data.
- **Frontend Software Engineer Intern, Onset Data Loggers – Bourne, MA** (May 2023 - August 2023)
 - Contributed to next-gen platform overhaul, delivering 2 key features (notification interface and item card sorting) to customers.
 - Partnered with middleware engineers to improve API efficiency, measurably reducing UI latency for end users.
 - Executed multiple user stories per sprint in Agile workflows, consistently accelerating iteration cycles and release cadence.
- **Founder & Content Creator, Independent Art Business** (January 2020 - December 2023)
 - Grew a digital art brand to 170k+ followers and millions of views through systematic, data-driven content experimentation.
 - Managed end-to-end physical product pipeline, selling 300+ units and generating \$1.6k profit within months.
 - Secured and executed official brand partnerships with Huion, XPPen, and Artisul across multi-platform marketing campaigns.

RECENT PROJECTS:

- **Ninja Mouse (Custom 3D Game Engine)**
 - Developed a C++ 3D game engine with component-based architecture, triangle-based collision detection, and AI pathfinding.
 - Abstracted engine components into 17 modules with OOP principles to allow scalability and robust per-game customization.
 - Integrated custom 3D assets with material loading support and environment shaders for enhanced visual fidelity.
 - Leveraged BVH and grid-based acceleration trees to logarithmically optimize environment creation and collision detection.
- **Origin Two (Custom 2D Game Engine)**
 - Developed Java-based 2D game engine with collision detection, AI pathfinding, animations, and dynamic level loading.
 - Implemented a precise, boundary-based physics system and custom UI framework for seamless player interaction.
 - Scaled engine to support 3+ dungeon crawler and platformer games alongside integrated sound and text generation.
- **Realtime Ocean Water Simulation**
 - Implemented a physically accurate ocean water simulation based on Jerry Tessendorf's paper on motion-picture oceans.
 - Built a fast fourier transform optimizer to allow real-time user manipulation based on wind speed and turbulence meters.
 - Introduced new features ocean foam, particle systems for spray, caustics, and custom shaders for reflection and refraction.
- **Realtime Graphics & Ray Tracing**
 - Created a real-time C++/OpenGL scene viewer with shape and camera manipulation and custom GLSL shaders.
 - Integrated ray tracing algorithm for realistic rendering using the Phong model, mirror reflections, and texture mapping.
 - Programmed custom shaders with framebuffer objects to achieve stylized post-processing effects.
- **3D Modeling & Animation**
 - Animated and imported game-ready and cyclable attack sequences with a custom 3D rig and character into Unreal.
 - Designed, modeled, rigged, textured, and animated 2 original characters with custom shaders for an animated short.
 - Created adaptable shader and texture customization sliders, supporting rapid creation of varying environments.
 - Led storyboarding, character design, and character modeling for a 3 minute animated short produced within 1 month.

SKILLS:

- **Computer Languages:** C++, Python, Java, React.js, Vue.js, Node.js, JS, TS, Shell, HTML/CSS, OpenGL, TensorFlow, Scipy
- **Tools:** Unreal, Maya, Blender, Gaea, Adobe, Git, Docker, Figma,
- **Skills:** Atlassian/Jira, Agile, CI/CD, Scrum, Kanban, Project Management